

2015 GSA Annual Meeting in Baltimore, Maryland, USA (1-4 November 2015)

Paper No. 330-13

Presentation Time: 4:50 PM

INTEGRATING SUSTAINABILITY AND GEOLOGY IN A NEW CURRICULUM FOR UNDERGRADUATE EARTH SCIENCE EDUCATION

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The accelerating pace of human industry and activity is causing profound changes to the function of earth systems that jeopardize the sustainability of human civilization. Unfortunately, earth science education at colleges and universities has been slow to recognize and respond to the challenges of anthropogenic modification of earth systems and most geoscience departments continue to teach a traditional curriculum that has been in place for generations. At Hofstra University we have integrated our programs in sustainability and geology and we are in the process of revising our curricula to emphasize eight themes that are fundamental to emerging issues of planetary sustainability:

1. Global Climate Change and Its Impact on Surface Processes and Environments
2. Erosion, Sedimentation, and Soil Formation
3. Mineral and Energy Resources
4. Water Resources
5. Geochemistry and Geochemical Cycles
6. Land Use, Land Use Change, Geoengineering, and Human Agency
7. Biosphere-Geosphere Interactions
8. Conservation Paleobiology

We have mapped these eight themes to the courses supporting our existing undergraduate degrees in Geology, Environmental Resources, and Sustainability and we are incorporating survey data from alumni and other professional stakeholders to determine how best to modify our existing curriculum to meet the demands of the 21st Century. Research integrating geology and sustainability is supported by the Hofstra University Stem Collaboratorium Initiative (HUSCI), which is a multi-million dollar project funded by the State of New York and Hofstra University that will bring together researchers in geology, sustainability, urban ecology, molecular biology, as well as environmental chemistry and environmental engineering to study global change and problems relating to environmental degradation, human health, urban habitats, and suburban development. Sustainability and earth systems are also addressed in our introductory courses and in educational outreach when discussing both local and global issues of climate change, water resources, and pollution with K-12 teachers, students, and the public.

Session No. 330

[T27. Intersections of Sustainability and Geosciences](#)

Wednesday, 4 November 2015: 1:30 PM-5:30 PM

Room 339 (Baltimore Convention Center)

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